

Opportunity OS

Full Schema, Logic, Workflow, and Feature Specification

Version 1.0 — April 2026

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Workflow, and Feature Specification

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1. Executive Overview

What Opportunity OS Is

Opportunity OS is a mobile-first CRM platform designed for relationship-driven sales teams operating in Healthcare and Government Contracting (GovCon) verticals. Unlike generic CRMs, it is purpose-built around the relationship intelligence and organizational hierarchy patterns specific to these markets.

The app is built as an Expo React Native application (iOS, Android, and Web), backed by an Express API server and a PostgreSQL database managed via Drizzle ORM.

Core Operating Model

Opportunity OS follows a dual-tier intelligence model:

- **Workspace tier:** Each client organization gets a private workspace. All CRM data — contacts, organizations, pipelines, opportunities, notes, tasks — lives here. This is the user's daily working environment.
- **Master Database tier:** Canonical reference data curated and maintained by the Opportunity OS platform team. Includes verified master organizations, master contacts, healthcare CMS overlays, GovCon identifiers, and relationship graph data. Workspace data can be "promoted" to the Master Database after admin review.

This separation ensures that client data remains isolated while the platform accumulates a growing intelligence layer that benefits all workspaces.

Admin vs Client Workspace Model

Platform Admin	Opportunity OS staff	Full system access via isPlatformAdmin flag; manages master DB, AI suggestions.
Workspace Owner/Admin	Client team leads	Manage their workspace: invite members, configure pipelines, approve business cards.
Workspace Member	Client sales reps	Read/write CRM data within their workspace; cannot manage workspace settings

The platform admin console is a separate, authenticated section of the same Expo app (route prefix /admin/), protected by platformAdminMiddleware.ts.

Master Database vs Workspace Database Relationship

```
%  
  
% [User submits org/contact]  
  
% [System enqueues to promotion queue]  
  
%%  
  
Master Promotion Queue (PENDING)  
  
%  
  
% [Platform admin reviews]  
  
% [Approve as NEW / MERGE / LINK / REJECT]  
  
%%  
  
Master Database (canonical, cross-workspace)  
  
%  
  
% [AI suggestions generated]  
  
% [Admin approves/rejects field suggestions]  
  
%%  
  
Master Org AI Suggestions (enrichment overlays)
```

Workspace organizations can be linked to master organizations via `masterOrganizationId`. This link is set by the platform admin after promotion review, not automatically by the system.

2. Full Schema

2.1 Enums Reference

Identity / Auth Enums	
workspace_role	OWNER, ADMIN, MEMBER
Organization Enums	
organization_type	HOSPITAL, HEALTH_SYSTEM, HOSPICE, HOME_HEALTH, GOVERNMENT_AGENCY, PRIME_CONTRACTOR,
organization_level	enterprise, group, facility
account_structure_type	enterprise, parent, regional, local_entity, facility
org_vertical	healthcare, govcon, general_business, government, nonprofit, vendor, other
primary_decision_level	enterprise, parent, regional, local
hierarchy_source_type	MASTER_DATABASE, EXTERNAL_ENRICHMENT, LLM_SYNTHESIS, HUMAN_CONFIRMED
Contact Enums	
contact_status	NEW, REVIEWED, ACTIVE, INACTIVE
stakeholder_role	DECISION_MAKER, INFLUENCER, CHAMPION, BLOCKER, OTHER
influence_level	LOW, MEDIUM, HIGH
relationship_strength_label	COLD, DEVELOPING, STRONG, STRATEGIC
Opportunity Enums	
opportunity_status	OPEN, WON, LOST, ON_HOLD
opportunity_vertical	HEALTHCARE, GOVCON, CONSULTING, PARTNERSHIP
Task / Activity Enums	
task_priority	LOW, MEDIUM, HIGH
task_status	OPEN, IN_PROGRESS, COMPLETED, CANCELED
activity_type	CALL, EMAIL, MEETING, CARD_SCAN, NOTE, FOLLOW_UP, EVENT, INTRO, LOGO_SCAN, ORG_ENRICHMENT, STRUCTURE_SCAN_STARTED, STRUCTURE_SUGGESTED, STRUCTURE_APPROVED, STRUCTURE_REJECTED

Business Card Enums

	Values
card_processing_status	UPLOADED, PARSING, PARSED, FAILED
card_review_status	PENDING_REVIEW, APPROVED, REJECTED, MERGED

Organization Scan Enums

org_scan_processing_status	UPLOADED, PARSING, PARSED, MATCHED, FAILED
org_scan_review_status	PENDING_REVIEW, APPROVED, REJECTED
admin_org_scan_processing_status	UPLOADED, PARSING, PARSED, MATCHED, FAILED
admin_org_scan_review_status	PENDING_REVIEW, APPROVED, REJECTED

Pipeline Enums

pipeline_view_template_status	draft, active, inactive, archived
-------------------------------	-----------------------------------

Master Organization Enums

master_relationship_type	SUBSIDIARY, REGIONAL, DBA, AFFILIATED
master_relationship_review_status	PENDING_REVIEW, APPROVED, REJECTED
master_org_industry	HEALTHCARE, GOVCON, GENERAL_BUSINESS
master_account_structure_type	ENTERPRISE, REGIONAL, FACILITY, SUB_FACILITY, GENERAL_ORG
master_validation_status	UNVALIDATED, PARTIALLY_VALIDATED, VALIDATED, REQUIRES_REVIEW
master_alias_type	DBA, ACQUIRED_BRAND, ABBREVIATION, FORMER_NAME, VARIANT
master_org_ai_suggestion_status	PENDING, APPROVED, REJECTED

Master Contact Enums

master_contact_role	DECISION_MAKER, INFLUENCER, CHAMPION, BLOCKER, OTHER
master_contact_influence	LOW, MEDIUM, HIGH
master_contact_validation_status	UNVALIDATED, VALIDATED, REQUIRES_REVIEW
promotion_entity_type	ORG, CONTACT, NOTE
promotion_change_type	CREATED, UPDATED, NOTE_ADDED
promotion_status	PENDING, APPROVED_NEW, APPROVED_MERGE, APPROVED_LINK, REJECTED

Onboarding Enums

	Values
onboarding_session_status	DRAFT, INTAKE, AWAITING_RECOMMENDATION, NORMALIZING, REVIEW, LOCKED, PROVISIONING,
onboarding_client_type	SINGLE_USER, SMALL_TEAM, ENTERPRISE
provisioning_step_key	CREATE_WORKSPACE, ASSIGN_PLAN, CREATE_MEMBERSHIPS, APPLY_VERTICAL_CONFIG,
provisioning_step_status	PENDING, IN_PROGRESS, COMPLETED, FAILED, SKIPPED
launch_checklist_item_status	SEED_TAGS, SEED_SAVED_VIEWS, SEED_DEFAULT_TASKS, PENDING, COMPLETED, SKIPPED, SEED_ALERTS, CREATE_LAUNCH_CHECKLIST, SEND_INVITE_EMAILS, RECORD_AUDIT_ENTRY,
workspace_add_on_status	ACTIVE, SUSPENDED, PENDING_CONFIG
onboarding_review_item_status	PENDING, APPROVED, EDITED, REJECTED
ai_confidence_band	HIGH, MEDIUM, LOW

Structure Scan Enums

structure_scan_status	PENDING, MASTER_MATCHED, EXTERNAL_SEARCHED, LLM_REVIEWED, COMPLETED, FAILED
structure_review_status	PENDING_REVIEW, APPROVED, REJECTED

Healthcare Intelligence Enums

cms_verification_status_enum	MATCHED, VERIFIED, NEEDS_REVIEW, REJECTED, IMPORT_ERROR
pain_point_category	ED_BOARDING, DISCHARGE_BOTTLENECK, CARE_TRANSITION_RISK, STAFFING_PRESSURE, CAPACITY_CONSTRAINT, REVENUE_CYCLE, CRITICAL_HIGH_MEDIUM_LOW
pain_point_severity	DOCUMENTATION_BURDEN, PATIENT_EXPERIENCE, OTHER
pain_point_frequency	CONSTANT, FREQUENT, OCCASIONAL, RARE
pain_point_source_type	CMS_SIGNAL, USER_REPORTED, ADMIN_CONFIRMED, ONBOARDING_EXTRACTED, CORROBORATING_SOURCE
evidence_type	QUANTITATIVE, QUALITATIVE, ANECDOTAL, INFERRED
pain_point_verification_status	SUGGESTED, PENDING_REVIEW, VERIFIED, REJECTED
competitor_type	INCUMBENT_VENDOR, EMERGING_VENDOR, INTERNAL_SOLUTION, MANUAL_PROCESS, NO_SOLUTION, UNKNOWN
incumbent_status	CONFIRMED_INCUMBENT, SUSPECTED_INCUMBENT, FORMER_INCUMBENT, NOT_INCUMBENT
contract_status	ACTIVE_CONTRACT, MONTH_TO_MONTH, EXPIRED, UNKNOWN
displacement_difficulty	VERY_HIGH, HIGH, MEDIUM, LOW
competitor_pain_point_relationship_type	CAUSED_BY, EXACERBATED_BY, MASKED_BY, OPPORTUNITY_ANGLE

2.2 Tables by Domain

Domain: Identity

users

OPPORTUNITY OS — FULL SPECIFICATION

		Required		Notes
id	text (UUID)	Yes	crypto.randomUUID()	PK
first_name	text	No	—	
last_name	text	No	—	
email	text	Yes	—	Unique; normalized to lowercase
password_hash	text	No	—	bcrypt hash
auth_provider_id	text	No	—	Unique; for OAuth providers
account_type	text	Yes	client_user	
is_platform_admin	boolean	Yes	false	Guards all /admin/ routes
platform_role	text	No	—	Future: sub-roles within platform team
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	Auto-updated

Domain: Workspace

id	text (UUID)	Yes	crypto.randomUUID()	PK
name	text	Yes	—	Workspace/company name
industry_focus	text	No	—	Free text; default on signup: "Healthcare & Pharma"
owner_user_id	text (FK!'users)	Yes	—	ON DELETE CASCADE
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	
Workspace Admin				
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE
user_id	text (FK!'users)	Yes	—	ON DELETE CASCADE
role	workspace_role	Yes	MEMBER	OWNER, ADMIN, MEMBER
created_at	timestamp	Yes	now()	

workspace_admin_audit_log

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OPPORTUNITY OS — FULL SPECIFICATION				
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!workspaces)	Yes	—	ON DELETE CASCADE
changed_by_user_id	text (FK!users)	No	—	ON DELETE SET NULL
changed_at	timestamp	Yes	now()	
action	text	Yes	—	e.g., INVITE_SENT, ROLE_CHANGED,
entity_type	text	Yes	—	e.g., workspace_member
entity_id	text	Yes	—	
previous_value	jsonb	No	—	Before state
new_value	jsonb	No	—	After state
platform_support_action	boolean	Yes	false	True when action taken by platform team
notes	text	No	—	Admin notes

Domain: CRM — Organizations

organizations

OPPORTUNITY OS — FULL SPECIFICATION

id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE
parent_organization_id	text (FK!'organizations)	No	—	Self-referencing; ON DELETE SET NULL
ultimate_parent_organization_id	text (FK!'organizations)	No	—	Denormalized; computed from parent chain
organization_level	organization_level	No	facility	enterprise, group, facility
account_structure_type	account_structure_type	No	—	enterprise, parent, regional, local entity, facility
vertical	org_vertical	No	—	healthcare, govcon, general business, etc.
primary_decision_level	primary_decision_level	No	—	Where buying decisions happen
name	text	Yes	—	Display name
legal_name	text	No	—	Full legal entity name
website	text	No	—	
phone	text	No	—	
email	text	No	—	
organization_type	organization_type	Yes	OTHER	
industry	text	No	—	Free text industry
sub_industry	text	No	—	
sub_vertical	text	No	—	
region_name	text	No	—	
msa_status	text	No	—	Master Service Agreement status
system_priority_tier	text	No	—	
expansion_strategy	text	No	—	
expansion_maturity	text	No	—	
strategic_tier	text	No	—	
address_line1	text	No	—	
address_line2	text	No	—	
city	text	No	—	
state	text	No	—	
zip	text	No	—	
country	text	No	—	
notes_text	text	No	—	Inline notes (not same as notes table)
owner_user_id	text (FK!'users)	No	—	Account owner
outreach_owner_user_id	text (FK!'users)	No	—	Who handles outreach
google_place_id	text	No	—	From logo/org scan enrichment
formatted_address	text	No	—	From Google Places
website_domain	text	No	—	Normalized domain for dedup

latitude	double precision	No	—	
longitude	double precision	No	—	
place_category	text	No	—	From Google Places
last_enriched_at	timestamp	No	—	When data was last enriched
enrichment_source	text	No	—	Which source enriched it
master_organization_id	text	No	—	Link to master organizations (no
hierarchy_confidence_score	double precision	No	—	0–1 score from hierarchy scan
hierarchy_last_scanned_at	timestamp	No	—	
hierarchy_last_reviewed_at	timestamp	No	—	
hierarchy_source_type	hierarchy_source_type	No	—	How the hierarchy was determined
suggested_parent_name	text	No	—	AI-suggested parent before review
suggested_ultimate_parent_name	text	No	—	AI-suggested root before review
onboarding_vertical_id	text (FK!'verticals)	No	—	From onboarding config
onboarding_sub_vertical_id	text (FK!'sub_verticals)	No	—	
organization_intelligence_summary	jsonb	No	—	Cached result of compute-intelligence-summary
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

Hierarchy business rules:

- parent_organization_id must belong to the same workspace
- Cycle detection runs before any parent assignment
- ultimate_parent_organization_id is computed by walking the parent chain
- When a parent changes, ultimate_parent_organization_id is propagated to all descendants (depth limit: 10)

organization_ems_profiles

One-to-one overlay for EMS/interfacility transport context on an organization.

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OPPORTUNITY OS — FULL SPECIFICATION				Notes
id	text (UUID)	Yes	auto	PK
organization_id	text (FK! organizations)	Yes	—	ON DELETE CASCADE
workspace_id	text (FK! workspaces)	Yes	—	ON DELETE CASCADE
primary_transport_need	text	No	—	
incumbent_provider	text	No	—	Current EMS provider
estimated_monthly_transports	integer	No	—	
payer_mix_summary	text	No	—	
las_vegas_jurisdiction_eligibility	text	No	—	City of Las Vegas jurisdiction flag
discharge_workflow_notes	text	No	—	
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

Domain: CRM — Contacts

contacts

OPPORTUNITY OS — FULL SPECIFICATION				Notes	Page 13
id	text (UUID)	Yes	auto	PK	
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE	
organization_id	text (FK!'organizations)	No	—	ON DELETE SET NULL	
first_name	text	No	—		
last_name	text	No	—		
full_name	text	Yes	—	Required; may be synthesized from first+last	
title	text	No	—	Job title	
department	text	No	—		
email	text	No	—		
phone	text	No	—		
mobile	text	No	—		
linkedin_url	text	No	—		
source	text	No	—	business_card, manual, import, etc.	
source_detail	text	No	—	Additional provenance info	
status	contact_status	Yes	NEW	NEW, REVIEWED, ACTIVE, INACTIVE	
notes_text	text	No	—	Inline notes	
owner_user_id	text (FK!'users)	No	—	ON DELETE SET NULL	
stakeholder_role	stakeholder_role	No	—	Buyer role in decision process	
influence_level	influence_level	No	—		
relationship_strength	integer	No	—	0–100 numeric score	
relationship_strength_label	relationship_strength_label	No	—	COLD, DEVELOPING, STRONG, STRATEGIC	
is_primary_relationship	boolean	Yes	false	Primary contact for org	
role_notes	text	No	—	Notes on their decision-making role	
master_contact_id	text (FK!'master_contacts)	No	—	Link to master DB contact	
created_at	timestamp	Yes	now()		
updated_at	timestamp	Yes	now()		

Domain: CRM — Opportunities and Pipelines

pipelines

Opportunity CS — Full Specification			Fields		Page 14
id	text (UUID)	Yes	auto	PK	
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE	
name	text	Yes	—		
category	text	No	—	e.g., EMS for EMS-specific pipelines	
created_at	timestamp	Yes	now()		
id	text (UUID)	Yes	auto	PK	
pipeline_id	text (FK!'pipelines)	Yes	—	ON DELETE CASCADE	
name	text	Yes	—	Stage display name	
stage_order	integer	Yes	—	Unique per pipeline	
probability_percent	integer	Yes	0	Win probability at this stage (0–100)	
created_at	timestamp	Yes	now()		

Unique constraint: (pipeline_id, stage_order)

opportunities

OPPORTUNITY OS — FULL SPECIFICATION					Notes
id	text (UUID)	Yes	auto	PK	
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE	
pipeline_id	text (FK!'pipelines)	Yes	—	ON DELETE CASCADE	
pipeline_stage_id	text (FK!'pipeline_stages)	Yes	—	ON DELETE RESTRICT (prevents stage deletion if opps exist)	
organization_id	text (FK!'organizations)	No	—	ON DELETE SET NULL	
primary_contact_id	text (FK!'contacts)	No	—	ON DELETE SET NULL	
title	text	Yes	—		
description	text	No	—		
vertical	opportunity_vertical	Yes	CONSULTING		
value_estimate	double precision	No	—	USD deal value	
close_date_estimate	timestamp	No	—		
status	opportunity_status	Yes	OPEN	OPEN, WON, LOST, ON_HOLD	
score	integer	No	—	Internal score (0–100)	
source	text	No	—	How the opp was sourced	
owner_user_id	text (FK!'users)	No	—	ON DELETE SET NULL	
stage_entered_at	timestamp	No	—	When the opp entered current stage	
service_line_id	text (FK!'service_lines)	No	—	ON DELETE SET NULL	
onboarding_vertical_id	text (FK!'verticals)	No	—		
onboarding_sub_vertical_id	text (FK!'sub_verticals)	No	—		
created_at	timestamp	Yes	now()		
updated_at	timestamp	Yes	now()		
opportunity_contacts					
id	text (UUID)	Yes	auto	PK	
opportunity_id	text (FK!'opportunities)	Yes	—	ON DELETE CASCADE	
contact_id	text (FK!'contacts)	Yes	—	ON DELETE CASCADE	
relationship_role	text	No	—	Their role on this specific deal	

Unique constraint: (opportunity_id, contact_id)

opportunity_ems_interfacility_profiles

One-to-one overlay for EMS opportunities.

OPPORTUNITY OS — FULL SPECIFICATION				Notes
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!workspaces)	Yes	—	
opportunity_id	text (FK!opportunities)	Yes	—	ON DELETE CASCADE; UNIQUE
service_mix_bls	boolean	Yes	false	Basic Life Support
service_mix_als	boolean	Yes	false	Advanced Life Support
service_mix_cct	boolean	Yes	false	Critical Care Transport
current_provider_name	text	No	—	Who currently holds the contract
estimated_monthly_transports	integer	No	—	
payer_mix_medicare_percent	integer	No	—	0–100
payer_mix_medicaid_percent	integer	No	—	0–100
payer_mix_private_percent	integer	No	—	0–100
payer_mix_other_percent	integer	No	—	0–100
primary_pain_points	text	No	—	Free text
agreement_status	text	No	—	e.g., PENDING, SIGNED
protocol_go_live_date	timestamp	No	—	
active_consistency_start_date	timestamp	No	—	When regular transport started
active_last_qualified_transport_at	timestamp	No	—	
qualified_transports_last_30_days	integer	No	—	
avg_qualified_transports_per_week	numeric	No	—	
jurisdiction_eligibility	text	No	—	
jurisdiction_notes	text	No	—	
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

pipeline_view_templates

Admin-managed templates for pipeline views. Published to workspaces during provisioning.

OPPORTUNITY OS — FULL SPECIFICATION

id

text (UUID)

Yes

auto

PK

Notes

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key	text	Yes	—	Unique slug
name	text	Yes	—	Display name
vertical	text	Yes	—	Which vertical this is for
sub_vertical	text	No	—	
status	pipeline_view_template_status	Yes	draft	draft, active, inactive, archived
is_locked	boolean	Yes	false	Admin-locked templates cannot be edited by
is_client_editable	boolean	Yes	true	Whether client can modify
config_json	jsonb	Yes	{}	Template configuration
created_by_user_id	text (FK!users)	No	—	
updated_by_user_id	text (FK!users)	No	—	
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

workspace_pipeline_views / workspace_pipeline_view_permissions

Workspace-specific instances of pipeline view templates with per-user permission overrides.

Domain: CRM — Tasks, Activities, Notes

tasks

OPPORTUNITY OS — FULL SPECIFICATION				Notes
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE
contact_id	text (FK!'contacts)	No	—	ON DELETE SET NULL
organization_id	text (FK!'organizations)	No	—	ON DELETE SET NULL
opportunity_id	text	No	—	No FK constraint (loose reference)
title	text	Yes	—	
description	text	No	—	
due_date	timestamp	No	—	
priority	task_priority	Yes	MEDIUM	
status	task_status	Yes	OPEN	
assigned_to_user_id	text (FK!'users)	No	—	ON DELETE SET NULL
created_by_user_id	text (FK!'users)	No	—	ON DELETE SET NULL
completed_at	timestamp	No	—	Set when status != COMPLETED
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE
contact_id	text (FK!'contacts)	No	—	ON DELETE SET NULL
organization_id	text (FK!'organizations)	No	—	ON DELETE SET NULL
opportunity_id	text	No	—	Loose reference
type	activity_type	Yes	—	See enum above
subject	text	Yes	—	
description	text	No	—	
occurred_at	timestamp	Yes	now()	
created_by_user_id	text (FK!'users)	No	—	ON DELETE SET NULL
created_at	timestamp	Yes	now()	

notes

OPPORTUNITY OS — FULL SPECIFICATION				Notes
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE
contact_id	text (FK!'contacts)	No	—	ON DELETE SET NULL
organization_id	text (FK!'organizations)	No	—	ON DELETE SET NULL
opportunity_id	text	No	—	Loose reference
content	text	Yes	—	Markdown or plain text
created_by_user_id	text (FK!'users)	No	—	ON DELETE SET NULL
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE
user_id	text (FK!'users)	No	—	ON DELETE SET NULL
entity_type	text	Yes	—	e.g., contact, organization
entity_id	text	Yes	—	
action	text	Yes	—	e.g., CREATE, UPDATE, DELETE
before_json	jsonb	No	—	Previous state
after_json	jsonb	No	—	New state
created_at	timestamp	Yes	now()	

Note: `audit_logs` is written by the schema but is not yet exposed via any API endpoint. Status: Planned.

Domain: CRM — Tags				
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE
name	text	Yes	—	Unique per workspace
color	text	No	—	Hex color
category	text	No	—	Tag grouping
created_at	timestamp	Yes	now()	

Unique constraint: (workspace_id, name)

contact_tags / organization_tags

Junction tables for many-to-many tag assignments. Each has (contact_id/organization_id, tag_id) unique constraint.

Domain: Business Card Scanner				
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE
uploaded_by_user_id	text (FK!'users)	No	—	ON DELETE SET NULL
image_url_front	text	Yes	—	GCS object path
image_url_back	text	No	—	GCS object path
raw_ocr_text	text	No	—	Raw text from OCR
parsed_json	jsonb	No	—	Structured fields from LLM parsing
processing_status	card_processing_status	Yes	UPLOADED	UPLOADED!'PARSING!'PARSED/FAILED
review_status	card_review_status	Yes	PENDING_REVIEW	PENDING_REVIEW!'APPROVED/REJECTED/
linked_contact_id	text (FK!'contacts)	No	—	Set when approved and contact created/linked
linked_organization_id	text (FK!'organizations)	No	—	Set when approved and org linked
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

Domain: Organization Logo Scans

organization_scans

OPPORTUNITY OS — FULL SPECIFICATION					Notes	Page 21
id	text (UUID)	Yes	auto	PK		
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE		
uploaded_by_user_id	text (FK!'users)	No	—	ON DELETE SET NULL		
organization_id	text (FK!'organizations)	No	—	ON DELETE SET NULL; which org is being identified		
image_url	text	Yes	—	US object path		
raw_ocr_text	text	No	—			
parsed_business_name	text	No	—	Extracted business name		
confidence_score	double precision	No	—	Match confidence (0–1)		
matched_place_json	jsonb	No	—	Array of Google Places matches		
selected_match_json	jsonb	No	—	Which match was selected		
processing_status	org_scan_processing_status	Yes	UPLOADED			
review_status	org_scan_review_status	Yes	PENDING_REVIEW			
created_at	timestamp	Yes	now()			
updated_at	timestamp	Yes	now()			
admin_org_scan_attempts						
id	text (UUID)	Yes	auto	PK		
Admin-side version for verifying master organizations via logs						
uploaded_by_admin_id	text (FK!'users)	No	—			
image_url	text	Yes	—			
raw_ocr_text	text	No	—			
parsed_business_name	text	No	—			
confidence_score	double precision	No	—			
matched_place_json	jsonb	No	—			
selected_match_json	jsonb	No	—			
processing_status	admin_org_scan_processing_status	Yes	UPLOADED			
review_status	admin_org_scan_review_status	Yes	PENDING_REVIEW			
created_master_org_id	text (FK!'master_organizations)	No	—	Set when approved and master org created		
created_at	timestamp	Yes	now()			
updated_at	timestamp	Yes	now()			

Domain: Organization Structure Scans

organization_structure_scans

Captures hierarchy inference results for a workspace org.

OPPORTUNITY OS — FULL SPECIFICATION					Notes
id	text (UUID)	Yes	auto	PK	
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE	
organization_id	text (FK!'organizations)	Yes	—	ON DELETE CASCADE	
initiated_by_user_id	text (FK!'users)	No	—	ON DELETE SET NULL	
scan_status	structure_scan_status	Yes	PENDING		Full pipeline: PENDING!'MASTER MATCHED!'EXTENDED
review_status	structure_review_status	Yes	PENDING_REVIEW		PENDING_REVIEW!'APPROVED/REJECTED/FAILED
suggested_parent_master_organization_id	text (FK!'master_organizations)	No	—		AI-suggested parent
suggested_parent_name	text	No	—		
suggested_ultimate_parent_name	text	No	—		
suggested_structure_type	text	No	—		
confidence_score	double precision	No	—		
evidence_summary	text	No	—		
external_source_payload	jsonb	No	—		Raw external API response
llm_reasoning_summary	text	No	—		LLM reasoning text
add_to_master_graph	boolean	Yes	false		Whether approved result should be written to
created_at	timestamp	Yes	now()		
updated_at	timestamp	Yes	now()		

Domain: Plans & Subscriptions				
id	text (UUID)	Yes	auto	PK
name	text	Yes	—	Display name
slug	text	Yes	—	Unique identifier (e.g., independent)
features	jsonb	No	—	Feature flags per plan
created_at	timestamp	Yes	now()	
id	text (UUID)	Yes	auto	PK
workspace_id	text (FK!'workspaces)	Yes	—	ON DELETE CASCADE
plan_id	text (FK!'plans)	Yes	—	No cascade; plan cannot be deleted if
status	text	Yes	active	active, cancelled, past due
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

Domain: Master Database

master_organizations

The canonical organizational reference record.

OPPORTUNITY OS — FULL SPECIFICATION				Notes
id	text (UUID)	Yes	auto	PK
canonical_name	text	Yes	—	Official name
display_name	text	No	—	Preferred display
normalized_name	text	Yes	—	Lowercased, punctuation-stripped for dedup
website_domain	text	No	—	
industry	master_org_industry	No	—	HEALTHCARE, GOVCON, GENERAL_BUSINESS
sub_vertical	text	No	—	
account_structure_type	master_account_structure_type	No	—	ENTERPRISE, REGIONAL, FACILITY, SUB-FACILITY, GENERAL_ORG
is_standalone	boolean	Yes	false	No Parent, Independent entity
confidence_score	double precision	Yes	0.5	Data quality score (0–1)
source_type	text	Yes	MANUAL	How it was created: MANUAL, WORKSPACE
source_confidence	double precision	Yes	1.0	Confidence in source
validation_status	master_validation_status	Yes	UNVALIDATED	UNVALIDATED, PARTIALLY_VALIDATED,
headquarters_address	text	No	—	
city	text	No	—	
state	text	No	—	
country	text	No	—	
notes	text	No	—	Admin notes
place_ids	jsonb (string[])	No	[]	Google Place IDs
aliases	jsonb (string[])	No	[]	Known alternative names
admin_flags	jsonb (string[])	No	[]	Admin annotation flags
structure_last_scanned_at	timestamp	No	—	
structure_last_reviewed_at	timestamp	No	—	
source_workspace_id	text (FK!'workspaces)	No	—	Which workspace promoted this
source_organization_id	text	No	—	The workspace org ID that was promoted
promoted_by_admin_user_id	text (FK!'users)	No	—	Which platform admin approved promotion
promoted_at	timestamp	No	—	
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

master_organization_aliases

OPPORTUNITY OS — FULL SPECIFICATION					Notes
id	text (UUID)	Yes	auto	PK	
master_organization_id	text (FK!'master_organizations)	Yes	—	ON DELETE CASCADE	
alias_name	text	Yes	—		
normalized_alias_name	text	Yes	—	Normalized for dedup	
alias_type	master_alias_type	Yes	VARIANT	DBA, ACQUIRED, BRAND,	
created_at	timestamp	Yes	now()		
				VARIANT	
master_organization_relationships					
id	text (UUID)	Yes	auto	PK	
parent_master_organization_id	text (FK!'master_organizations)	Yes	—	ON DELETE CASCADE	
child_master_organization_id	text (FK!'master_organizations)	Yes	—	ON DELETE CASCADE	
relationship_type	master_relationship_type	Yes	SUBSIDIARY	SUBSIDIARY, REGIONAL, DBA, AFFILIATED	
confidence_score	double precision	Yes	1.0		
evidence_summary	text	No	—	Why this relationship was established	
source_payload	jsonb	No	—	Raw data used to establish link	
approved_by_user_id	text (FK!'users)	No	—		
review_status	master_relationship_review_status	Yes	APPROVED	PENDING_REVIEW, APPROVED, REJECTED	
created_at	timestamp	Yes	now()		
updated_at	timestamp	Yes	now()		
master_org_healthcare_overlays					
id	text (UUID)	Yes	auto	PK	
master_organization_id	text (FK!'master_organizations)	Yes	—	UNIQUE; ON DELETE CASCADE	
facility_type	text	No	—	e.g., HOSPITAL, ASC, SNF	
licensed_beds	integer	No	—		
trauma_level	text	No	—	LEVEL_I–LEVEL_IV	
system_type	text	No	—	e.g., ACADEMIC_MEDICAL_CENTER, COMMUNITY_HOSPITAL	
ownership_model	text	No	—	FOR_PROFIT, NON_PROFIT,	
care_setting	text	No	—		
created_at	timestamp	Yes	now()		
updated_at	timestamp	Yes	now()		

master_org_govcon_overlays

One-to-one GovCon-specific data for a master org.

OPPORTUNITY OS — FULL SPECIFICATION				Default	Page 20
id	text (UUID)	Yes	auto	PK	
master_organization_id	text (FK!"master_organizations)	Yes	—	UNIQUE; ON DELETE CASCADE	
uei	text	No	—	Unique Entity Identifier (SAM.gov)	
cage_code	text	No	—		
naics_codes	jsonb (string[])	No	[]		
prime_or_sub	text	No	—		
contract_vehicles	jsonb (string[])	No	[]	e.g., GSA Schedule, GWAC	
agency_alignment	text	No	—	Primary federal agency	
created_at	timestamp	Yes	now()		
updated_at	timestamp	Yes	now()		

master_org_ai_suggestions				
id	text (UUID)	Yes	auto	PK
AI-generated field-level suggestions for master orgs, expires next				
master_organization_id	text (FK!'master_organizations)	Yes	—	ON DELETE CASCADE
field	text	Yes	—	Field name (e.g., industry)
current_value	text	No	—	Current field value
suggested_value	text	Yes	—	AI suggestion
rationale	text	No	—	Why the AI suggests this
status	master_org_ai_suggestion_status	Yes	PENDING	PENDING, APPROVED, REJECTED
reviewed_by_user_id	text (FK!'users)	No	—	
reviewed_at	timestamp	No	—	
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

Domain: Master Contacts and Promotion Queue

master_contacts

OPPORTUNITY OS — FULL SPECIFICATION				Notes
id	text (UUID)	Yes	auto	PK
master_organization_id	text (FK!'master_organizations)	Yes	—	ON DELETE CASCADE
full_name	text	Yes	—	
first_name	text	No	—	
last_name	text	No	—	
title	text	No	—	
department	text	No	—	
email	text	No	—	
phone	text	No	—	
mobile	text	No	—	
linkedin_url	text	No	—	
stakeholder_role	master_contact_role	No	—	
influence_level	master_contact_influence	No	—	
confidence_score	double precision	Yes	0.5	
validation_status	master_contact_validation_status	Yes	UNVALIDATED	
notes	text	No	—	
source_workspace_id	text (FK!'workspaces)	No	—	
source_contact_id	text	No	—	Workspace contact that was promoted
promoted_by_admin_user_id	text (FK!'users)	No	—	
promoted_at	timestamp	No	—	
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

master_promotion_queue

Holds pending promotion requests for admin review.

OPPORTUNITY OS — FULL SPECIFICATION			Default	Page 20
id	text (UUID)	Yes	auto	PK
entity_type	promotion_entity_type	Yes	—	ORG, CONTACT, NOTE
entity_id	text	Yes	—	ID within the workspace
workspace_id	text (FK!workspaces)	Yes	—	ON DELETE CASCADE
change_type	promotion_change_type	Yes	—	CREATED, UPDATED, NOTE_ADDED
status	promotion_status	Yes	PENDING	PENDING, APPROVED_NEW, APPROVED_MERGE, APPROVED_LINK, resolution
resolved_master_id	text	No	—	Master entity ID after resolution
rejection_reason	text	No	—	
source_snapshot	jsonb	No	—	Snapshot of workspace entity at time of
resolved_by_user_id	text (FK!users)	No	—	Platform admin who resolved
resolved_at	timestamp	No	—	
created_at	timestamp	Yes	now()	
updated_at	timestamp	Yes	now()	

Domain: Onboarding and Provisioning

verticals / sub verticals / service lines / add on types

Global configuration tables managed by the platform team. Define what verticals, service lines, and add-ons are available system-wide.

verticals	key (unique), label, is_active, sort_order	e.g., key=healthcare
sub_verticals	vertical_id, key (unique per vertical), label	e.g., key=ems under healthcare
service_lines	vertical_id, sub_vertical_id, key, default_pipeline_template_key	Links to pipeline templates
add_on_types	key (unique), label, config_schema	Feature add-ons (e.g., GovCon module)

workspace_onboarding_config

workspace_id	text (FK! workspaces)	Yes	—	UNIQUE
vertical_id	text (FK! verticals)	No	—	Resolved FK
sub_vertical_id	text (FK! sub_verticals)	No	—	
vertical_text	text	No	—	Legacy free-text backup
sub_vertical_text	text	No	—	Legacy
default_contact_roles	jsonb	Yes	[]	List of default contact role strings

workspace_service_lines

Which service lines are enabled for a workspace.

OPPORTUNITY OS — FULL SPECIFICATION

workspace_id + service_line_id	—	Yes	—	Unique pair
is_enabled	boolean	Yes	true	
custom_label	text	No	—	Client can rename
custom_config	jsonb	No	—	Per-workspace config overrides
enabled_by_admin_user_id	text (FK!users)	No	—	

workspace_add_ons

workspace_id + add_on_type_id	—	Yes	—	Unique pair
status	workspace_add_on_statuses	Yes	ACTIVE	ACTIVE, SUSPENDED, PENDING CONFIG
config	jsonb	Yes	{}	

client_onboarding_sessions

id	text (UUID)	Yes	auto	PK
One session per onboarding engagement (admin-led)				
status	onboarding_session_statuses	Yes	DRAFT	Full lifecycle: DRAFT!INTAKE!AWAITING_RECOMMENDATION!INTEGRATION!SMALL_TEAM
client_type	onboarding_client_types	Yes	SMALL_TEAM	SMALL_TEAM
intake_payload	jsonb	Yes	{}	Admin-entered client details
grok_raw_payload	jsonb	No	—	Raw AI response
grok_model_version	text	No	—	Which AI model was used
grok_confidence	double precision	No	—	Overall AI confidence
normalized_recommendation	jsonb	No	—	Normalized AI output after grokNormalizer.ts
admin_decisions	jsonb	Yes	{}	Per-item admin approve/edit/reject
applied_config	jsonb	No	—	Final config used for provisioning
created_from_preset_id	text	No	—	If started from a preset
created_workspace_id	text (FK!workspaces)	No	—	Set after CREATE_WORKSPACE step
created_by_admin_user_id	text (FK!users)	No	—	
notes	text	No	—	Admin notes
normalized_at / locked_at / provisioned_at / archived_at	timestamp	No	—	Lifecycle timestamps

onboarding_provisioning_steps

One row per provisioning step per session. Tracks progress.

OPPORTUNITY OS — FULL SPECIFICATION				Notes
workspace_id	text (FK! workspaces)	Yes	—	
snapshot_date	timestamp	Yes	now()	
setup_completeness_pct	integer	Yes	0	0–100
active_user_count	integer	Yes	0	
contact_count	integer	Yes	0	
org_count	integer	Yes	0	
opportunity_count	integer	Yes	0	
missing_data_flags	jsonb	Yes	[]	List of identified gaps
grok_improvement_suggestions	jsonb	Yes	[]	AI improvement suggestions
onboarding_presets				
name	text	Yes	—	
vertical_id + sub vertical id	text (FK)	No	—	
is_public	boolean	Yes	false	Available to all admins
preset_payload	jsonb	Yes	{}	Full intake+config payload
usage_count	integer	Yes	0	Auto-incremented
version	integer	Yes	1	
created_from_session_id	text	No	—	
workspace_intelligence				
workspace_id	text (FK! workspaces)	Yes	—	
kind	text	Yes	—	e.g., saved_view, alert, pattern
key	text	Yes	—	Unique per (workspace, kind)
label	text	Yes	—	
severity	text	No	—	For alerts
data	jsonb	Yes	{}	Content
source	text	Yes	onboarding	
is_active	boolean	Yes	true	

Unique constraint: (workspace_id, kind, key)

Domain: Healthcare Intelligence

organization_healthcare_profile

One-to-one with organizations. Stores CMS data with full traceability.

OPPORTUNITY OS — FULL SPECIFICATION				Notes
organization_id	text (FK! organizations)	Yes	Unread	UNIQUE
workspace_id	text (FK! workspaces)	Yes	—	
cms_ccn	text	No	—	CMS Certification Number
cms_provider_type	text	No	—	Free text (enum normalization deferred)
cms_ownership_type	text	No	—	
cms_bed_count	integer	No	—	Licensed beds
cms_emergency_services	boolean	No	—	
cms_overall_star_rating	integer	No	—	1–5
cms_patient_experience_rating	integer	No	—	1–5
cms_ed_total_time_minutes	integer	No	—	ED door-to-disposition
cms_ed_time_to_admit_minutes	integer	No	—	ED door-to-inpatient
cms_ed_boarding_time_minutes	integer	No	—	ED boarding time
cms_ed_lwbs_percent	integer	No	—	Left Without Being Seen; stored as basis points
cms_care_transition_rating	integer	No	—	
cms_patient_experience_subscores_json	jsonb	No	—	Record<string, number>
cms_raw_json	jsonb	No	—	Full raw CMS response
cms_source	text	No	—	
cms_verification_status	cms_verification_status_enum	No	—	MATCHED, VERIFIED, NEEDS REVIEW, REJECTED
cms_last_updated_at	timestamp	No	—	When CMS data was last refreshed
cms_source_url	text	No	—	IMPORT_ERROR
cms_dataset_name	text	No	—	e.g., Hospital Compare
cms_dataset_version	text	No	—	
cms_extracted_at	timestamp	No	—	When extracted from CMS
cms_effective_date	date	No	—	Data effective date
cms_match_method	text	No	—	ccn_exact, name_fuzzy, manual
cms_match_confidence_score	integer	No	—	0–100

Stale threshold: CMS data older than 90 days triggers amber warning in UI.

organization_pain_points

Pain points for healthcare orgs. CMS signals !' SUGGESTED only; promotion to VERIFIED requires admin approval.

OPPORTUNITY OS — FULL SPECIFICATION				Notes
organization_id	text (FK! organizations)	Yes	indexed	ON DELETE CASCADE
workspace_id	text (FK! workspaces)	Yes	—	
department	text	No	—	Affected clinical department
pain_point_category	pain_point_category	Yes	—	
pain_point_statement	text	No	—	Human-readable description
severity	pain_point_severity	Yes	MEDIUM	
frequency	pain_point_frequency	No	—	
source_type	pain_point_source_type	Yes	—	
source_reference	text	No	—	Which CMS metric or user input
evidence_type	evidence_type	No	—	
linked_cms_signal_key	text	No	—	e.g., cms_ed_boarding_time minutes
confidence_score	integer	Yes	50	0–100
verification_status	pain_point_verification_status	Yes	SUGGESTED	Never auto-VERIFIED; must be approved
is_active	boolean	Yes	true	Soft delete
reviewed_by_user_id	text (FK! users)	No	—	
reviewed_at	timestamp	No	—	
review_note	text	No	—	

Idempotency key (CMS suggestions): (org_id, category, linked_cms_signal_key, source_type) — prevents duplicate suggestions.

organization_competitors

OPPORTUNITY OS — FULL SPECIFICATION			Notes	Page 34
organization_id	text (FK! organizations)	Yes	—	
workspace_id	text (FK! workspaces)	Yes	—	
competitor_name	text	Yes	—	
competitor_type	competitor_type	Yes	UNKNOWN	
service_line	text	No	—	
incumbent_status	incumbent_status	Yes	NOT_INCUMBENT	
share_of_wallet_estimate	integer	No	—	0–100; null = unknown
contract_status	contract_status	No	UNKNOWN	
strengths	jsonb (string[])	No	[]	
weaknesses	jsonb (string[])	No	[]	
pain_points_caused	jsonb (string[])	No	[]	DERIVED/CACHED — do not write directly; re-
displacement_difficulty	displacement_difficulty	No	MEDIUM	
source_type	pain_point_source_type	No	—	
source_reference	text	No	—	
confidence_score	integer	Yes	50	
verification_status	pain_point_verification_status	Yes	SUGGESTED	
is_active	boolean	Yes	true	
competitor_pain_point links				
organization_competitor_id	text (FK! organization_competitors)	Yes	—	ON DELETE CASCADE
organization_pain_point_id	text (FK! organization_pain_points)	Yes	—	ON DELETE CASCADE
relationship_type	competitor_pain_point_relationship_type	Yes	CAUSED_BY	CAUSED_BY, EXACERBATED_BY,
confidence_score	integer	Yes	50	
notes	text	No	—	

When any link row is created or deleted, the API layer must call refreshPainPointsCausedCache(competitorId) to re-aggregate organization_competitors.pain_points_caused.

3. Per-Feature Logic

3.1 Authentication

Purpose: Authenticate users and associate them with a workspace.

Inputs: email, password, optional rememberMe (login); email, password, firstName, lastName, workspaceName (signup).

Outputs: JWT token containing { userId, workspaceId, email }.

Business Rules:

- Emails are normalized to lowercase on creation and lookup
- Passwords must be at least 6 characters
- Login finds workspace by owner lookup first, then falls back to workspace_members
- A user without a workspace cannot log in
- On signup: user + workspace + OWNER membership + "independent" plan subscription are created atomically

Validations:

- Email uniqueness checked before creation (HTTP 409 on conflict)
- Password length validation (min 6 chars)

Triggers / Automations:

- Workspace creation includes default industryFocus = "Healthcare & Government Contracting"
- "independent" plan assigned automatically; if plan not found in DB, subscription is skipped silently

Edge Cases:

- forgot-password endpoint is a stub — returns a success message without sending any email. Email delivery is Planned.
- OAuth/external auth provider field (auth_provider_id) exists in schema but no OAuth provider is wired up.
- A user who was invited (existing user added to workspace) logs in via their own workspace first; they cannot switch workspaces from the UI.

Role involved: None (pre-auth). After login: OWNER is assumed for self-signup users.

3.2 Business Card Scanner (OCR Pipeline)

Purpose: Convert a photographed business card into a verified contact + org record.

Inputs: Image file (JPEG/PNG, max 20 MB), uploaded via multipart/form-data.

Outputs: business_card record !' contact record + optional organization record on approval.

Business Rules:

1. Image is uploaded to Google Cloud Storage (GCS) via objectStorageClient
2. OCR is triggered via parseBusinessCardImage() (GPT-4o Vision)
3. Raw OCR text !' structured JSON (parsedJson) via LLM parsing

4. Card enters PENDING_REVIEW state; user reviews in the card review screen
5. On APPROVE: contact is created from parsed fields; if org name present, org is created or matched; card is marked APPROVED; activity log entry created
6. On REJECT: card is marked REJECTED; no entities created
7. On MERGE: card is linked to an existing contact; MERGED status

Validations:

- File must be present (HTTP 400 if missing)
- OCR availability check via isOcrAvailable() — if OCR fails, card stays in UPLOADED state

Side Effects:

- Contact creation triggers enqueuePromotion() to add to master promotion queue (if enabled)
- Activity of type CARD_SCAN is created

Approval Requirements: The reviewing user must be a workspace member. No admin-only gate on approval.

PHI Warning: PHI warning is displayed in the card review form when card contains health-related data.

Edge Cases:

- Back-of-card image is optional
- Parsed JSON may have missing fields; review UI shows blanks
- Duplicate contact detection is not implemented at approval time — Planned

3.3 Organization Logo/Brand Scan

Purpose: Identify an organization from its logo or signage photograph.

Inputs: Image file uploaded via scan interface.

Outputs: Organization matched to a Google Places result; org record enriched with place data.

Business Rules:

1. Image uploaded to GCS
2. OCR extracts business name text
3. Confidence-scored match against Google Places API
4. User reviews candidate matches and selects correct one
5. On APPROVE: workspace org is updated with google_place_id, formatted_address, latitude, longitude, place_category; activity of type LOGO_SCAN created

Validations: Organization must belong to caller's workspace.

Edge Cases:

- Multiple candidate matches returned as matched_place_json; user picks via selected_match_json
- Scan can be done without a pre-existing organization (creates new org if needed)

3.4 Organization Hierarchy and Structure Scans

Purpose: Infer parent/child/ultimate-parent relationships for a workspace organization using the master database, external sources, and LLM reasoning.

Inputs: organization_id to scan.

Outputs: organization_structure_scans record with suggested parent hierarchy; activity log entries.

Pipeline Stages:

1. PENDING !' Check master database for known parent
2. MASTER_MATCHED (if found) !' Suggest from master DB
3. EXTERNAL_SEARCHED !' Search external APIs for additional evidence
4. LLM_REVIEWED !' LLM synthesizes evidence and proposes hierarchy
5. COMPLETED / FAILED

Approval Flow:

- Result enters PENDING_REVIEW state
- On APPROVE: parent_organization_id and ultimate_parent_organization_id set on the org; hierarchy_source_type = LLM_SYNTHESIS or MASTER_DATABASE; activities logged as STRUCTURE_APPROVED
- On REJECT: activities logged as STRUCTURE_REJECTED; org hierarchy unchanged

If add_to_master_graph = true: Approved hierarchy is also written to master_organization_relationships table.

3.5 Master Database Promotion

Purpose: Elevate high-quality workspace org/contact data to the canonical master database.

Inputs: Workspace organization_id or contact_id; optional change type (CREATED, UPDATED, NOTE_ADDED).

Outputs: master_promotion_queue record; after admin resolution: master_organizations or master_contacts record.

Business Rules:

- Queue is appended to by enqueuePromotion() helper — called after org/contact creation or significant updates
- Admin resolves each queue item as:
 - APPROVED_NEW !' create new master record
 - APPROVED_MERGE !' merge data into existing master record
 - APPROVED_LINK !' link workspace record to existing master record without merging data
 - REJECTED !' discard

Validations:

- source_snapshot captures the entity state at time of promotion request (not the latest state)
- Admin cannot auto-promote; every promotion requires explicit human decision

Downstream Effects:

- On APPROVED_NEW or APPROVED_MERGE: master_organization_id or master_contact_id field on workspace entity is updated

- promoted_by_admin_user_id and promoted_at written to master record
- source_workspace_id and source_organization_id preserved for provenance

3.6 Master Org AI Suggestions

Purpose: Use AI (Grok/OpenAI) to suggest field-level enrichments for master organization records.

Inputs: master_organization_id; the AI analyses the org's current data and external signals.

Outputs: master_org_ai_suggestions records with PENDING status.

Business Rules:

- AI suggestions are NEVER auto-applied; every suggestion requires admin APPROVE or REJECT
- A normalization safety net (normalizeFieldValue()) maps AI output variations to valid enum values before storing
- Per-field suggestions: industry, accountStructureType, healthcare.facilityType, healthcare.traumaLevel, healthcare.systemType, healthcare.ownershipModel, healthcare.careSetting, govcon.primeOrSub, govcon.agencyAlignment, govcon.naicsCodes, etc.

Approval Flow:

- APPROVE: field value written to master_organizations or its overlay tables; suggestion marked APPROVED
- REJECT: suggestion marked REJECTED; field unchanged

Edge Cases:

- AI may produce values outside enum; normalizer maps to nearest valid value or preserves raw value
- Suggestions can be regenerated; new suggestions created but previous ones not overwritten

3.7 Account Intelligence Pulse (Org Intelligence)

Purpose: Compute a real-time relationship health + opportunity status summary for an organization.

Inputs: Organization ID and all associated contacts, activities, tasks, opportunities from workspace.

Outputs: OrgIntelligenceResult with: accountState, health score (0–100), risk score, coverageGaps, primaryAction, openOpportunities, contacts with computed strength.

Business Rules (rules-based, no AI):

Account state determination:

- COLD: no activity in last 30 days AND no open opportunities
- WARMING: activity in last 30 days, no open opportunities
- ACTIVE: activity in last 14 days, has open opportunities
- AT_RISK: has open opportunity with stage not progressed in 14+ days OR overdue task 14+ days old
- EXPANDING: multiple open opportunities with recent activity

Health score (0–100) based on:

- Activity recency within 14 days
- Number of primary contacts with relationship strength
- Open opportunity win probability

Risk score:

- Elevated by: overdue tasks, stale pipeline stages, inactive primary contacts

Coverage gaps:

- Missing DECISION_MAKER, CHAMPION, or other key stakeholder roles
- No primary relationship contact

Primary action:

- Computed from account state and most urgent signal (e.g., "Schedule follow-up", "Advance deal to next stage", "Capture missing contact")

Trigger: Called on demand via GET /api/organizations/:id/intelligence.

3.8 Healthcare Intelligence (CMS + Pain Points + Competitors)

Purpose: Provide healthcare-specific intelligence for accounts in the healthcare vertical.

Inputs: organization_id (must have vertical = 'healthcare').

Outputs: CMS profile data; verified and suggested pain points; competitor landscape; 7-dimension opportunity score; cached intelligence summary.

CMS Profile Logic:

- CMS data stored in organization_healthcare_profile
- cms_verification_status determines display: MATCHED/VERIFIED = green; NEEDS_REVIEW = amber
- Stale threshold: cms_last_updated_at > 90 days !' amber warning displayed

Pain Points Logic:

- CMS signals auto-generate SUGGESTED pain points (idempotent by (org_id, category, cms_signal_key, source_type))
- SUGGESTED !' PENDING_REVIEW !' VERIFIED (requires admin approve) or REJECTED
- VERIFIED pain points shown in "Verified" tab; SUGGESTED shown in "Suggested" tab
- "Needs Review" dot shown to ALL workspace members (not admin-gated) when SUGGESTED items exist

Competitor Logic:

- Competitors tracked with incumbent status, strengths, weaknesses, displacement difficulty
- pain_points_caused on competitors is DERIVED — re-aggregated from competitor_pain_point_links whenever links change (never written directly)
- Strengths displayed as green dot; weaknesses as red dot

7-Dimension Opportunity Score:

Dimensions: CMS data completeness, pain point severity, competitor weakness, buyer access, stakeholder coverage, recent activity, entry strategy clarity. Each dimension scored 0–100, weighted, combined into overall score. Thresholds: "e70 = green, 40–69 = amber, <40 = red.

Intelligence Summary:

- Cached in `organizations.organization_intelligence_summary` (jsonb)
 - Recomputed via `POST /organizations/:id/compute-intelligence-summary`
 - Shape: { `topPainPoints[]`, `topCompetitors[]`, `buyerPatterns[]`, `entryStrategy`, `primaryAction`, `impactStatement`, `computedAt` }
-

3.9 Admin-Led Onboarding and Provisioning

Purpose: Allow platform admins to configure and provision a new client workspace with AI-assisted setup.

Inputs (Intake): `clientName`, `website`, `industryDescription`, `productsSold`, `customerType`, `salesCycleType`, `teamSize`.

AI Recommendation (Grok/OpenAI):

- AI analyzes intake and recommends: vertical, sub-vertical, service lines, pipeline templates, contact roles, add-on modules
- Confidence band (HIGH/MEDIUM/LOW) shown per item
- All items enter PENDING status as `onboarding_review_items`

Review Phase:

- Admin approves, edits, or rejects each item
- Session cannot lock until all `is_required = true` items are resolved
- Once all resolved, session status != LOCKED; `applied_config` generated

Provisioning Phase (16 sequential steps):

	Description
CREATE_WORKSPACE	Create workspace record
ASSIGN_PLAN	Assign subscription plan
CREATE_MEMBERSHIPS	Create owner membership
APPLY_VERTICAL_CONFIG	Set vertical/sub-vertical config
ENABLE_SERVICE_LINES	Enable selected service lines
ENABLE_ADD_ONS	Enable selected add-ons
PUBLISH_PIPELINE_TEMPLATES	Clone pipeline templates !' workspace pipelines
SEED_CONTACT_ROLES	Seed default contact roles
SEED_TAGS	Seed default tags
SEED_SAVED_VIEWS	Seed default saved views
SEED_DEFAULT_TASKS	Seed default onboarding tasks
SEED_ALERTS	Seed intelligence alerts
CREATE_LAUNCH_CHECKLIST	Create launch checklist items by client type
SEND_INVITE_EMAILS	Send invite emails — STUB: not implemented
RECORD_AUDIT_ENTRY	Write audit log
SNAPSHOT_HEALTH_BASELINE	Take initial workspace health snapshot

Launch Checklists by Client Type:

- SINGLE_USER: REVIEW_PIPELINE, ADD_FIRST_CONTACT, CONFIRM_FIRST_TARGET
- SMALL_TEAM: + ASSIGN_USERS, CONFIGURE_PERMISSIONS
- ENTERPRISE: + CONFIGURE_REPORTING

Retry Logic: Failed steps can be retried via retryFailedOnly = true flag without re-running completed steps.

4. Full Workflow Map

4.1 Self-Service Signup

1. User visits public landing page
2. User taps "Sign Up" !' signup screen
3. User enters: first name, last name, email, password, workspace name
4. Backend validates: email format, password length "e 6, email uniqueness
5. Backend creates: users record, workspaces record (industry_focus = "Healthcare & Government Contracting"), workspace_members record (role = OWNER), subscriptions record (plan = "independent")
6. JWT token returned; user logged in as OWNER
7. User redirected to dashboard (empty state)
8. No email verification. No guided first-run. No industry selection.

4.2 Admin-Led Onboarding

1. Platform admin logs in via /admin/login (requires is_platform_admin = true)
2. Admin navigates to /admin/onboarding/new
3. Admin fills intake form (client details, sales cycle type, team size)
4. System creates client_onboarding_sessions record in INTAKE status
5. Admin triggers AI recommendation !' session !' AWAITING_RECOMMENDATION
6. AI (Grok) analyzes intake !' returns recommended config !' normalizedAt set; session !' REVIEW
7. Admin reviews each onboarding_review_items line: approve / edit / reject
8. All required items resolved !' session !' LOCKED; appliedConfig computed
9. Admin triggers provisioning !' PROVISIONING
10. 16 provisioning steps execute sequentially with retry on failure
11. Session !' PROVISIONED; created_workspace_id set
12. Admin navigates to launch screen
13. Launch screen shows activation summary (pipeline count, view count, task count)
14. Admin taps "Open Priority Dashboard" to enter workspace

4.3 Organization Creation and Updates

1. User taps "Add Organization" or "New Organization"
2. User enters: name (required), type, website, phone, address fields, vertical, tags
3. Backend validates: name is non-empty, workspace membership
4. Organization created in workspace
5. If parent_organization_id provided: cycle detection runs; ultimate_parent_organization_id computed; parent org must be in same workspace
6. If parent changes: propagateUltimateParent() updates all descendants
7. enqueuePromotion() called with changeType = CREATED for master DB review
8. Tags applied via organization_tags junction

Updates:

1. PATCH request with changed fields
2. If parent changes: cycle check !' ultimate parent recomputed !' propagated to descendants
3. enqueuePromotion() called with changeType = UPDATED
4. updatedAt auto-updated

4.4 Contact Creation and Updates

1. User adds contact manually or via business card approval
2. Required: full_name (or derived from first+last)
3. Optional: org link, title, department, email, phone, LinkedIn, status, tags, stakeholder role, relationship strength
4. On creation: status = NEW; source set to creation method
5. If linked to org: shows on org detail screen's contacts tab
6. enqueuePromotion() called for master DB review

From Business Card Approval:

1. Card review screen shows parsed fields
2. User can edit any field before confirming
3. On approve: contact created with source = business_card; org optionally created/linked; card review_status = APPROVED; linked_contact_id set

4.5 Business Card Intake !' OCR !' Review !' Approval

1. User opens scanner !' camera or gallery
2. Image uploaded to GCS via POST /api/business-cards/upload
3. Card record created with processing_status = UPLOADED
4. OCR triggered: parseBusinessCardImage() via GPT-4o Vision
5. processing_status !' PARSING !' PARSED (or FAILED)
6. parsed_json contains: { name, title, company, email, phone, mobile, linkedin, address, notes }
7. Card enters PENDING_REVIEW queue
8. User sees card in review list; taps to open review screen
9. User reviews/edits extracted fields
10. User approves !' contact created !' card review_status = APPROVED
11. User rejects !' card review_status = REJECTED
12. User merges !' linked to existing contact !' card review_status = MERGED

4.6 Master Database Sync (Promotion Workflow)

1. Workspace org or contact is created/updated
2. enqueuePromotion() inserts into master_promotion_queue with status = PENDING and source_snapshot of entity

3. Platform admin views promotion queue at /admin/master-promotion/queue
4. Admin sees: entity type, workspace name, change type, snapshot data
5. Admin checks for existing master record match (name/domain similarity)
6. Admin decides:
 - APPROVED_NEW: Creates new master_organizations or master_contacts; sets workspace entity's master_organization_id or master_contact_id
 - APPROVED_MERGE: Merges snapshot data into existing master record; links workspace entity
 - APPROVED_LINK: Links workspace entity to existing master without changing master data
 - REJECTED: Queue item closed; no action
7. Queue item updated with status, resolved_master_id, resolved_by_user_id, resolved_at

4.7 Duplicate Detection and Merge Handling

Current State: Partial implementation.

- Org dedup: normalized_name computed via normalizeOrgName() (lowercased, punctuation-stripped); website_domain normalized via normalizeDomain()
- Contact dedup: Not implemented in workspace layer; email uniqueness not enforced within workspace
- On promotion to master DB: admin manually checks for matches; master_organization_aliases used for known variants
- Automated merge UI: Planned

4.8 Notes Handling

1. User adds note on contact detail, org detail, or opportunity detail screen
2. Note linked to up to three entities via contact_id, organization_id, opportunity_id (all optional/nullable)
3. Notes support plain text (markdown not enforced by backend)
4. Create, Read, Update, Delete all implemented
5. Notes appear in the "Notes" tab on each detail screen
6. Notes with significant intelligence content can be included in promotion queue (entity_type = NOTE, change_type = NOTE_ADDED)

4.9 Pain Point Capture !' Suggestion !' Approval

1. CMS data is loaded for an org (organization_healthcare_profile)
2. CMS signal engine analyzes CMS metrics:
 - cms_ed_boarding_time_minutes > threshold !' generates ED_BOARDING pain point suggestion
 - cms_ed_lwbs_percent > threshold !' generates CARE_TRANSITION_RISK suggestion
 - cms_overall_star_rating "d 2 !" generates PATIENT_EXPERIENCE suggestion
 - etc.
3. Suggestion created as organization_pain_points with verification_status = SUGGESTED, source_type = CMS_SIGNAL; is_active = false

4. Idempotency check prevents duplicate suggestions for same (org_id, category, linked_cms_signal_key, source_type)
5. "Needs Review" dot shown on PainPointsCard for all workspace members
6. Workspace admin reviews: clicks Approve !' verification_status = VERIFIED, is_active = true
7. Or Reject !' verification_status = REJECTED
8. Verified pain points shown in "Verified" tab; used in opportunity scoring and intelligence summary

4.10 Competitor Tracking

1. Admin or workspace member adds competitor entry to an org
2. Fields: name, type, service line, incumbent status, strengths (array), weaknesses (array), displacement difficulty, contract status, share of wallet
3. verification_status defaults to SUGGESTED; workspace admin can verify
4. Links to pain points created via competitor_pain_point_links (relationship type: CAUSED_BY, EXACERBATED_BY, etc.)
5. When links change: refreshPainPointsCausedCache() re-aggregates pain_points_caused on competitor
6. Competitor accordion in mobile UI: collapsed shows name + type + incumbent badge; expanded shows strengths (green dot), weaknesses (red dot), linked pain points

4.11 Opportunity Creation and Pipeline Movement

1. User taps "Add Opportunity"
2. Required: title, pipeline, pipeline stage
3. Optional: organization, primary contact, value estimate, close date, description, service line
4. Opportunity created with status = OPEN
5. stage_entered_at set on creation and on each stage change
6. Pipeline Kanban board: opportunities displayed as cards in columns by stage
7. User drags card to new column !' pipeline_stage_id updated; stage_entered_at reset
8. Status changes: WON, LOST, ON_HOLD set via status field update

EMS-specific:

- If pipeline.category = 'EMS': EMS saved views strip shown; EMS Transport Profile card on opportunity detail
- EMS profile can be created/updated with service mix, transport data, payer mix, agreement info

4.12 Follow-up Task Generation

1. User manually creates task from contact detail, task list, or org detail
2. Fields: title (required), description, due date, priority, assignment, linked contact/org/opportunity
3. Tasks appear on global task list with status and due-date filters
4. Overdue detection: due_date < now() and status = OPEN
5. Automated task generation: Planned — no AI-generated tasks yet

4.13 Reporting and Rollups

- 1. GET /api/reports aggregates workspace data:
 - Contact count by status
 - Opportunity count and value by stage/status
 - Activity count by type
 - Task completion rate
- 2. Roll-up stats on org hierarchy: contacts, opportunities, pipeline value, won value across all child orgs (recursive)
- 3. workspace_health_snapshots captures point-in-time workspace metrics
- 4. Advanced reporting dashboards: Planned

4.14 Admin Review Flows

- 1. Master Org AI Suggestions: Admin views pending suggestions at /admin/master-orgs/:id/suggestions; approves/rejects each field
- 2. Promotion Queue: Admin reviews pending promotions at /admin/master-promotion/queue
- 3. Onboarding Review: Admin reviews AI recommendations per item in onboarding session
- 4. Business Card Review: Workspace admins/members review cards
- 5. Pain Point Review: Workspace admins approve/reject suggested pain points
- 6. Structure Scan Review: Workspace admins approve/reject hierarchy suggestions

4.15 Archive / Deactivate / Restore Behavior

Organizations	is_active not in schema; soft delete not implemented; hard delete cascades contacts/opps
Contacts	status = INACTIVE serves as soft deactivation
Pain Points	is_active = false for rejected/inactive; still exists in DB
Competitors	is_active = false for inactive
Onboarding Sessions	archived_at timestamp set; not deleted
Business Cards	Status changes to REJECTED; card record retained

5. Per-Feature Specification

Authentication

Purpose	Authenticate users, manage sessions
Schema Entities	users, workspaces, workspace_members, subscriptions, plans
Workflows	Self-service signup, login
Role Permissions	All roles: login/logout. OWNER implied on self-signup
Status	Complete (core); Planned (forgot password email, email verification)
What's Built	Login, signup, /auth/me, change-password
Missing Logic	Forgot-password email not sent; no email verification; no OAuth
Missing UX	No "verify your email" screen; forgot-password shows success but does nothing

Organization Management

Status	Complete
What's Built	Full CRUD, hierarchy (parent/child), 11 saved views, tag system, roll-up stats, vertical-specific display labels
Missing Logic	No soft delete; duplicate detection within workspace not implemented
Missing UX	No bulk edit; no dedup merge UI

Contact Management

Status	Complete
What's Built	Full CRUD, tags, org linking, stakeholder roles, relationship strength
Missing Logic	No duplicate detection; no email uniqueness enforcement within workspace
Missing UX	No contact merge UI

Business Card Scanner

Detail	
Status	Complete
What's Built	Upload, OCR (GPT-4o), review screen, approve/reject/merge, PHI warning
Missing Logic	No duplicate detection against existing contacts at approval time
Missing UX	No batch scan; no bulk review

Pipeline / Kanban

Status	Complete
What's Built	Horizontal Kanban board, multiple pipelines, stage probability, EMS saved views
Missing Logic	Stage change automation (auto-tasks) not implemented
Missing UX	No pipeline analytics view

Opportunities

Status	Complete
What's Built	Full CRUD, contacts many-to-many, EMS profile overlay, pipeline stage tracking
Missing Logic	No automated stage-based actions
Missing UX	No forecast/reporting view

Tasks

Status	Complete (list + CRUD); In Progress (detail/edit screen)
What's Built	Create, complete, filter by status/priority/due, link to contact/org/opp
Missing Logic	No AI task generation; no reminders/push notifications
Missing UX	No dedicated task edit screen; no task calendar view

Notes



Detail	
Status	Complete
What's Built	CRUD on contact/org/opp detail screens
Missing Logic	No rich text; no @mention; no note-to-task conversion

Admin-Led Onboarding

Status	Complete (provisioning); Planned (email handoff, client portal)
What's Built	Full 5-phase flow: intake !' AI recommend !' review !' provision !' launch
Missing Logic	SEND_INVITE_EMAILS step is a stub
Missing UX	No client-facing progress view; no admin re-provisioning UI

Master Database

Status	Complete (core); In Progress (AI suggestion workflows)
What's Built	Master orgs + contacts, promotion queue, AI suggestions, relationship graph, overlays
Missing Logic	No automated dedup; admin must manually match
Missing UX	No visual relationship graph; no org family tree view

Healthcare Intelligence

Status	Complete (schema + API + mobile cards)
What's Built	CMS profile, pain points (suggest/verify), competitor tracking, 7-dimension scoring, intelligence summary
Missing Logic	CMS data import pipeline not built (data entered manually or via API); no automated refresh
Missing UX	CMS data entry via admin portal not built

EMS Vertical

Status	Complete
What's Built	EMS pipeline, opportunity profile (22 columns), organization profile, saved views, jurisdiction tracking
Missing Logic	No reporting specific to EMS transport metrics

6. Permissions and Control Layers

Tier 1: Platform Admin (is_platform_admin = true)

Enforced by platformAdminMiddleware.ts on all /admin/* routes.	
Access master database	Full CRUD on master_organizations, master_contacts, overlays, aliases, relationships
Manage promotion queue	Approve/reject/merge all pending workspace promotions
Manage AI suggestions	Approve/reject field-level suggestions for master orgs
Run onboarding sessions	Full access to all client onboarding sessions
Manage pipeline templates	Create/edit/delete pipeline_view_templates
Manage onboarding presets	Create/edit/delete onboarding_presets
View all workspaces	List and inspect any workspace
Run org scans (admin)	Logo scan to create new master orgs
View platform stats	Dashboard metrics across all workspaces
Cannot access	Individual workspace CRM data directly (no bypass of workspace auth for standard routes)

Tier 2: Workspace Owner (role = OWNER)

All ADMIN capabilities	Full access
Cannot be demoted if last admin	System prevents removing last OWNER/ADMIN
Invite and remove members	Full team management

Tier 3: Workspace Admin (role = ADMIN)

Enforced by workspace role check in route handlers. In healthcare intelligence routes, checked via isAdminCaller().

	Details
Approve/reject business cards	Yes
Approve/reject pain points	Yes (ADMIN or OWNER)
Approve/reject structure scans	Yes
Approve/reject competitors	Yes
Manage workspace pipeline views	Yes
Cannot	Manage plans, access master DB admin routes

Tier 4: Workspace Member (role = MEMBER)

Create/read/update organizations	Yes
Create/read/update contacts	Yes
Create/read/update opportunities	Yes
Create/read/update tasks, activities, notes	Yes
Upload and OCR business cards	Yes
View pain points (both tabs)	Yes — "Needs Review" dot visible to all
Approve/reject pain points	No — ADMIN/OWNER only
Access admin console	No
Invite members	No

Middleware Enforcement

authMiddleware.ts	Verifies JWT; injects userId, workspaceId on all /api/* routes
platformAdminMiddleware.ts	Verifies is_platform_admin = true; guards all /admin/* routes
getCurrentWorkspace()	Used in route handlers to validate workspace membership
isAdminCaller()	Used in healthcare intelligence routes to check OWNER/ADMIN role

7. Master Database Intelligence Model

What Data Can Flow from Workspace to Master DB		
		Who Controls
Organizations	Enqueued by enqueuePromotion() after create/update	Platform admin resolves
Contacts	Enqueued by enqueuePromotion() after create/update	Platform admin resolves
Notes	Enqueued manually as NOTE_ADDED change type	Platform admin resolves
Business card data	Indirectly via contact creation	Goes through standard contact flow
Healthcare intelligence	Not promoted to master DB (workspace-scoped only)	N/A

What Auto-Suggests Only

Master org field enrichments	AI generates master_org_ai_suggestions with PENDING status
Pain points (healthcare)	CMS signals auto-create SUGGESTED pain points in workspace
Hierarchy suggestions	Structure scans create PENDING_REVIEW suggestions

Nothing is ever auto-applied to the master database. All promotions and suggestion approvals require explicit human action.

What Requires Approval

Workspace org != master org (new)	Platform admin
Workspace org != master org (merge)	Platform admin
Workspace org != link to existing master	Platform admin
Workspace contact != master contact	Platform admin
AI suggestion != apply to master org field	Platform admin
CMS pain point suggestion != verified	Workspace OWNER/ADMIN
Organization hierarchy suggestion != apply	Workspace OWNER/ADMIN

What Should Never Auto-Promote

- Personal data (PII) — contacts should always be manually reviewed
- Pain points — CMS signals are only evidence, not ground truth
- Hierarchy relationships — require evidence synthesis

Validation Pipeline Rules

- 1. Name Normalization: normalizeOrgName() strips punctuation, lowercases, collapses whitespace
- 2. Domain Normalization: normalizeDomain() extracts root domain for dedup
- 3. Confidence Scoring: master_organizations.confidence_score reflects overall data quality (0–1)
- 4. Validation Status Progression: UNVALIDATED !' PARTIALLY_VALIDATED !' VALIDATED (manual admin action)
- 5. AI Field Normalization: normalizeFieldValue() maps AI-produced variations to valid enum values

Provenance/Source Tracking

Every master record tracks:

- source_type — how it was created (MANUAL, WORKSPACE_PROMOTED, ADMIN_SCAN)
- source_workspace_id — which workspace contributed it
- source_organization_id — workspace entity ID
- promoted_by_admin_user_id — who approved it
- promoted_at — when

Confidence Scoring

0.0–0.3	Unvalidated, single source, sparse data	
0.3–0.6	Partially validated, some corroboration	
0.6–0.9	Validated, multiple sources	
0.9–1.0	Fully validated, confirmed from authoritative source	
Completeness scoring (master org):		
Canonical Name	15	Yes
Website Domain	15	Yes
Industry	10	Yes
Account Structure Type	10	Yes
Normalized Name	5	No
Validation Status (not UNVALIDATED)	10	No
+ Healthcare/GovCon overlay fields	5–10 each	Vertical-only

Health Stages: INCOMPLETE (0–39%), IDENTIFIED (40–59%), STRUCTURED (60–79%), STRATEGIC (80–100%).

Merge and Duplicate Governance

- Before promotion: Admin checks for existing master org by name similarity and domain
- Alias table: master_organization_aliases stores known variants; checked during dedup

- Merge action: Admin selects APPROVED_MERGE; data from workspace snapshot applied where master field is empty
 - No automated merge: No fuzzy-match auto-merge; human decision required
-

8. UX Logic

8.1 Authentication Screens

		Admin Screens	Behind Scenes
Login	Email + password form, "Remember me", "Forgot"	Same	JWT stored in SecureStore (native) or localStorage (web)
Signup	Name, email, password, workspace name fields	Same	User + workspace created; OWNER role assigned
Forgot Password	Email input ! "If account exists, reset link sent"	Same	Stub: no email sent
Logged In	Dashboard with workspace data	Admin console accessible at /admin/	/auth/me fetched on app load
Token Expired	Redirected to login	Same	Token verification fails; token cleared

8.2 Organization Detail Screen

Normal	Org header, intel pulse, hierarchy, contacts, pipeline	Same + admin-only actions	Data fetched from multiple endpoints
No contacts	Empty state with "Add Contact" CTA	Same	GET /organizations/:id returns contacts: []
No opportunities	Empty pipeline summary	Same	Opportunities list empty
Healthcare org	CMS Evidence Card, Pain Points Card (2 tabs), Competitor Landscape, Entry Strategy	Same + Approve/Reject pain points	Healthcare endpoints fetched conditionally
Stale CMS data	Amber warning on CMS card Intelligence Pulse with Opp	Same	cms_last_updated_at > 90 days triggers warning
Needs Review (pain points)	Red dot on Pain Points tab header	Same	Any SUGGESTED items exist

8.3 Business Card Review Screen

Processing	Loading indicator	OCR in progress
Parsed	Extracted fields editable	parsedJson shown
PHI Warning	Yellow warning banner	If health-related data detected
Approved	Confirmation; contact created	review_status = APPROVED
Failed OCR	Error message; fields empty	processing_status = FAILED

8.4 Pain Points Card

Verified tab	List of verified pain points with severity badges	Same + Approve/Reject buttons on Suggested tab
Suggested tab	List of suggested pain points with CMS trigger shown	Approve/Reject buttons per item
Empty (verified)	"No verified pain points" empty state	+ CTA to review suggestions
Needs Review dot	Visible to ALL workspace members	Same

8.5 Competitor Landscape Card

Widget States	
Collapsed	Competitor name, type badge, incumbent badge, displacement pill
Expanded	Strengths (green dot), weaknesses (red dot), linked pain points count
Empty	"No competitors tracked" with admin CTA to add

8.6 Admin Onboarding Screens

Intake	Normal	Form with client details
Recommend	Loading	"Analyzing client profile..." spinner
Review	Items pending	List of AI-suggested items with confidence badges
Review	Item approved	Green checkmark
Review	Item edited	Edit indicator; admin's value shown
Review	Session locked	"Lock & Provision" CTA enabled
Provision	Running	Step list with status icons (pending/in progress/completed/failed)
Provision	Step failed	Red error; retry option
Launch	Done	Activation summary; "Open Priority Dashboard" CTA

8.7 Master DB Admin Screens

Promotion Queue	Pending items	Entity snapshot, workspace name, change type
Promotion Queue	Resolved item	Status badge (Approved/Rejected/Merged)
AI Suggestions	Pending	Field name, current value, suggested value, rationale
AI Suggestions	Approved	Green checkmark; value written to master org
Completeness	Any org	Score bar, health stage, missing fields list, next best action

9. Open Gaps / Risks / Needed Decisions

Unresolved Logic Gaps

		Priority
forgot-password endpoint is a stub	Users cannot reset passwords; support blocker	HIGH
SEND_INVITE_EMAILS provisioning step is a stub	Invited team members have no way to know they have access	HIGH
No email delivery infrastructure	Password reset, invites, and welcome emails all blocked	HIGH
No invite acceptance flow for new users	Inviting unknown email does nothing useful	HIGH
No duplicate contact detection at business card approval	Duplicate contacts created silently	HIGH
Self-serve users have no industry/vertical selection	All workspaces default to "Healthcare & GovCon" regardless of user type	MEDIUM
audit_logs table exists but no API exposes it	Audit data not queryable or visible	MEDIUM
opportunity_id on tasks/notes/activities has no FK constraint	Referential integrity not enforced; can reference deleted opps	MEDIUM
Automated CMS data import pipeline not built	Healthcare intelligence CMS data must be manually entered via API	HIGH
No automated pain point refresh when CMS data updates	Old CMS data keeps old suggestions; no refresh trigger	MEDIUM

Unresolved UX Gaps

No first-run wizard for self-serve users	Empty dashboard; no guidance	HIGH
No Day-1 onboarding checklist for self-serve users	Only admin-provisioned workspaces get a launch checklist	HIGH
No client-facing progress view during admin onboarding	Client has no visibility into setup status	MEDIUM
No automated handoff email when admin provisioning completes	Client has no credentials or welcome info	HIGH
No task edit/detail screen	Users can create and complete tasks but cannot edit details inline	MEDIUM
No pipeline analytics/forecast view	No revenue forecasting UI	MEDIUM
No duplicate merge UI	Admins cannot merge duplicate contacts or orgs in the UI	MEDIUM
No visual org relationship graph	Master org hierarchy not visualized	LOW
No push notifications or reminders	Users have no overdue task alerts	MEDIUM

Unresolved Schema Gaps

	Notes
users.platform_role not enforced	Field exists but no business logic uses it
organizations.msa_status, expansion_strategy, expansion_maturity not validated	Free text; no picklist enforcement
tasks.opportunity_id lacks FK	Loose reference; can go stale
Plan features field is jsonb but not validated	No schema for which feature keys are valid
workspace_intelligence.kind values not enumerated	Any string accepted; no type enforcement

Technical Risks

No email delivery service	CRITICAL	Without email: password reset impossible, invites broken, no notifications
No dedup at scale	HIGH	As workspace data grows, duplicate org/contact records will accumulate
GCS dependency for all media	HIGH	Business cards, org scans — all depend on GCS availability
API TypeScript errors pre-existing	MEDIUM	esbuild ignores TS errors at runtime; type safety not enforced in API layer
JWT tokens long-lived with no refresh	MEDIUM	Tokens expire but there's no silent refresh; users get logged out unexpectedly
No rate limiting on AI endpoints	MEDIUM	AI suggestion generation has no rate limit; could incur unexpected cost
as any forbidden but TS types may drift	LOW	Codebase rule; requires ongoing enforcement

Recommended Next Build Priorities

1	Email delivery integration (e.g., Resend, SendGrid)	Unblocks: password reset, invites, welcome emails
2	Invite acceptance flow for new users	Unblocks: team onboarding from admin provisioning
3	Self-serve first-run wizard	Reduces churn from empty-state users
4	Duplicate contact detection at business card approval	Prevents data quality issues
5	CMS data import pipeline	Required for healthcare intelligence to be useful at scale
6	Vertical/industry selection at self-serve signup	Makes onboarding relevant to each user
7	Automated pain point refresh on CMS update	Keeps intelligence current
8	Task edit/detail screen	Completes task management feature
9	Pipeline analytics/forecast view	Sales leadership need
10	Push notifications for overdue tasks	User retention and engagement

10. Appendix: Feature Inventory Matrix



OPPORTUNITY OS — FULL SPECIFICATION

User Login	JWT-based login with	users, workspaces,	Login flow	All	Complete	—	—	—
User Signup (Self-Service)	Creates user + workspace	users, workspaces, workspace members, su	Self-service signup	None (pre-auth)	Complete	No email verification:	No guided onboarding	HIGH
Forgot Password	OWNER membership reset request	workspace members, su	—	All	Planned	No email selection; no	Shows success	HIGH
Change Password	Authenticated password change	users	—	All	Complete	—	—	—
Admin Login	Platform admin authn	users	—	Platform Admin only	Complete	—	—	—
Organization CRUD	Create/read/update/delete/child/ultimate	organizations, organization tags, tags	Org creation, updates	All (create/update); OWNER/ADMIN	Complete	No soft delete; no dedup	No bulk edit; no merge UI	MEDIUM
Org Hierarchy	11 pre-built filter views	organizations	—	All	Complete	—	—	—
Org Saved Views	Create/read/update/delete/	contacts, contact tags, contacts	Contact creation, BC	All	Complete	No dedup	No merge UI	MEDIUM
Contact CRUD	Stakeholder role, contacts, camera, address, OCR	business_cards, contacts,	Business card flow	All (scan/review);	Complete	No dedup at approval	No batch scan	HIGH
Contact Relationship Fields	Photo !' Google	organization_scans,	Logo scan flow	All	Complete	—	—	LOW
Business Card	Photo !' Google	organization_scans,	Logo scan flow	All	Complete	—	—	LOW
Org Logo Scan	Photo !' Google	organization_scans,	Logo scan flow	All	Complete	—	—	LOW
Org Structure	Horizontal board	pipelines, pipeline stages,	Opp creation, stage movement	All	Complete	No stage automation	No analytics	MEDIUM
Pipeline (Kanban)	grouped by workspace	opportunities	—	All	Complete	—	—	—
Multiple Pipelines	EMS-specific 8-stage pipeline	pipelines, opportunity ems	EMS-specific flow	All	Complete	—	—	—
EMS Pipeline	Create/read/update/delete/	opportunities, opportunity	Opp creation,	All	Complete	—	No forecast view	MEDIUM
Opportunity CRUD	22-column EMS	opportunity_ems	—	All	Complete	—	—	LOW
EMS Opportunity Profile	transport profile	profiles	Follow-up task flow	All	Complete (list); In	No AI generation;	No edit screen; no	HIGH
Tasks	Activity feed	activities	—	All	Complete	—	—	—
Activities	Activity feed	activities	—	All	Complete	—	—	—
Notes	Note CRUD on any entity	notes	Notes flow	All	Complete	No rich text	—	LOW
Tags	Workspace-scoped tags	tags, contact tags, organi	—	All	Complete	—	—	—
Account Intelligence	Rules-based AI/ML	tags, contact tags, organi	—	All	Complete	No AI; no trend history	—	MEDIUM
Admin-Led Onboarding	5-phase AI-assisted	client_onboarding_session	Admin onboarding flow	Platform Admin	Complete (core);	Email not sent; no client portal	No re-provisioning UI	HIGH
Onboarding Presets	workspace provisioning	onboarding_review_item	—	Platform Admin	Planned (email)	—	—	LOW
Master Org DB	Canonical org	master_organizations, master_*	Master DB sync	Platform Admin	Complete	No automated dedup	No graph view	HIGH
Master Promotion	reference database	master_organizations, master_*	Promotion workflow	Platform Admin	Complete	Admin must manually	—	HIGH
Master Org AI	AI-generated field	master_org_ai_suggestions	AI suggestion flow	Platform Admin	Complete	No rate limiting	—	MEDIUM
Suggestions Master	enrichments	master_org_ai_suggestions	—	Platform Admin	Complete	—	—	MEDIUM
Contact DB	Canonical contact	contacts	Promotion workflow	Platform Admin	Complete	—	—	MEDIUM
Healthcare Intelligence	CMS data on healthcare	organization_healthcare_profiles	Healthcare intel flow	All (view); OWNER/ADMIN	Complete (schema + complete)	No import pipeline; no auto refresh	No CMS data entry admin UI	HIGH
Healthcare Intelligence	Suggested/verified pain	organization_pain_points	Pain point flow	All (view); OWNER/ADMIN	Complete	—	—	HIGH
Healthcare Intelligence	Competitor tracking with	organization_competitors,	Competitor tracking flow	All (view); OWNER/ADMIN	Complete	—	—	MEDIUM
Healthcare Intelligence — Opp Score	7-dimension opportunity scoring	competitor_pain_point_links, organization_pain_points, organization_compet	—	ADMIN (verify)	—	—	—	—

itors

		—	All	Complete	—	—	LOW
Healthcare Intelligence — Summary	Cached intelligence summary on						

org

organizations
.organization
intelligence

summary

—	All	Complete	Manual trigger only	—	MEDIUM
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Team Management /	Invite/remove	workspace_members, workspace_invites	Invite flow	OWNER/ADMIN	Complete (add existing)	No email for new-user	No invite acceptance	HIGH
Plan Management	Workspace subscription plans	plans, subscriptions (computed)	Signup	OWNER	Complete (data model); invite new (billing UI)	No payment integration	No upgrade/downgrade	HIGH
Reports	Workspace analytics		Reporting flow	All		No advanced	No charts/dashboards	MEDIUM
Workspace Health	Point-in-time workspace health	workspace_health_snapshots	Onboarding provisioning	Platform Admin	Complete	Not exposed to clients	No client-facing health dashboard	MEDIUM
Snapshots	Entity-level records	audit_logs, workspace_audit_logs	—	Platform Admin	In Progress	Audit log not exposed via UI	No audit trail	LOW
Public Landing Page	Marketing + pricing page	—	—	—	Complete	—	—	—

(generates)